

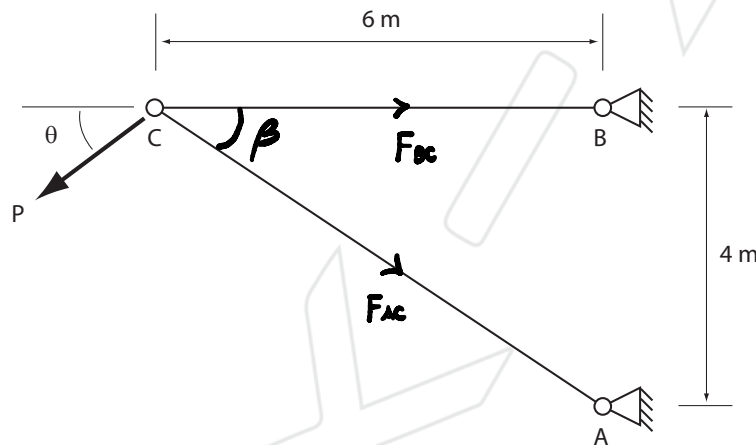
## AERO40008 – Structures

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## Equilibrium

(1.) At what angles ( $\theta$ ) should the force  $P$  be applied so that the magnitude of force in bar AC is 80% of the magnitude of the force in bar BC? N.B. there are four answers.

Determine the support reactions in terms of  $P$  and  $\theta$ .



At equilibrium,

$$(\rightarrow), P \cos \theta = + F_{BC} + F_{AC} \cos \beta = + F_{BC} \pm 80\% F_{BC} \frac{3}{13} \sqrt{13} = \frac{+65 \pm 12\sqrt{13}}{65} F_{BC}$$

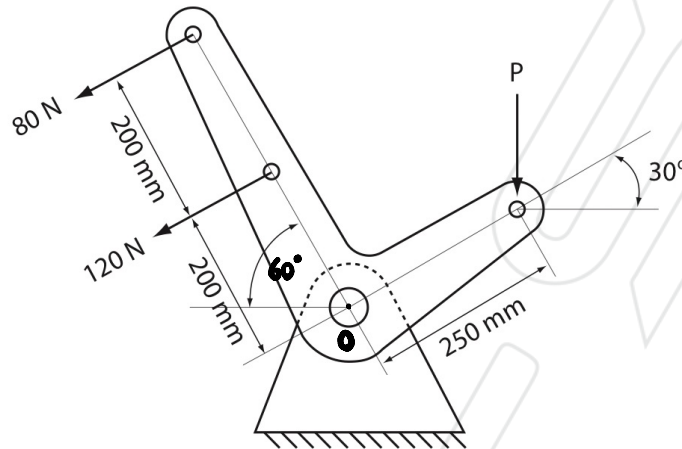
$$(\uparrow), P \sin \theta = - F_{AC} \sin \beta = \mp 80\% F_{BC} \frac{2}{13} \sqrt{13} = \mp \frac{8\sqrt{13}}{65} F_{BC}$$

$$\therefore \frac{+65 \pm 12\sqrt{13}}{65} \tan \theta = \mp \frac{8\sqrt{13}}{65}$$

$$\text{If } F_{AC} = 0.8 F_{BC}, \theta = 165.08^\circ \text{ OR } 345.08^\circ$$

$$\text{If } F_{AC} = -0.8 F_{BC}, \theta = 53.00^\circ \text{ OR } 233.00^\circ$$

(2.) The crank mechanism in the figure is shown in an equilibrium position. Determine the required force  $P$  acting on the crank, and the magnitude and direction of the resultant force at the bearing



$M(O),$

$$120\text{ N} \times 200\text{ mm} + 80\text{ N} \times (200\text{ mm} + 200\text{ mm}) = P \times (250\text{ mm} \times \cos 30^\circ)$$

$$\therefore P = 258.65\text{ N}$$

At bearing,

$$F(\rightarrow) = -(80\text{ N} + 120\text{ N}) \times \sin 60^\circ = -173.21\text{ N}$$

$$F(\uparrow) = -(80\text{ N} + 120\text{ N}) \times \cos 60^\circ - P = -358.65\text{ N}$$

$$\therefore F = 398.29\text{ N}$$

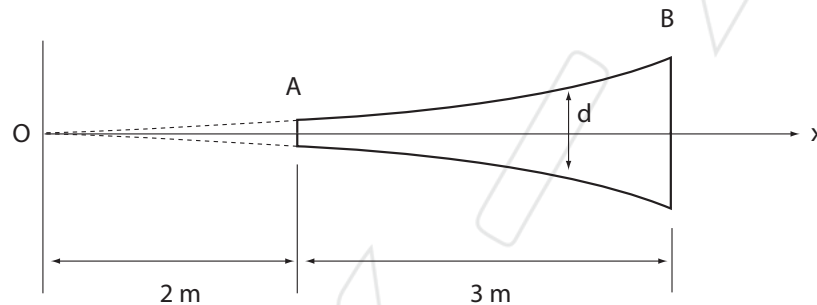
$$\theta(\rightarrow) = -115.78^\circ$$

## Stress and Strain

(3.) The bar AB is 3 m long and has a circular cross-section with variable diameter. This diameter is expressed by the relationship  $d = kx^2$ , in which  $x$  is measured from the origin  $O$ , and  $k$  is a constant. The bar is made from titanium alloy ( $E = 119 \text{ GPa}$ ) and the diameter at B is 30 mm.

The bar is subjected to an axial tensile load of 13.5 kN.

Determine a) the maximum stress in the bar, and b) the elongation of the bar if the elastic limit of the alloy is not exceeded i.e. the material does not yield.



Answer a) 746 MPa, b) 3.9 mm

$$(a) \text{ At B, } 30\text{mm} = k(5\text{m})^2 \quad \therefore k = \frac{3}{2500} \text{ m}^{-1} \quad \therefore d = \frac{3}{2500} x^2$$

$$\therefore d' = \frac{6}{2500} x \quad (2 \leq x \leq 5) \quad \therefore d_{\min} = 4.8 \text{ mm when } x = 2 \text{ m}$$

$$\therefore A_{\min} = \pi \left( \frac{1}{2} d_{\min} \right)^2 = 1.810 \times 10^{-5} \text{ m}^2$$

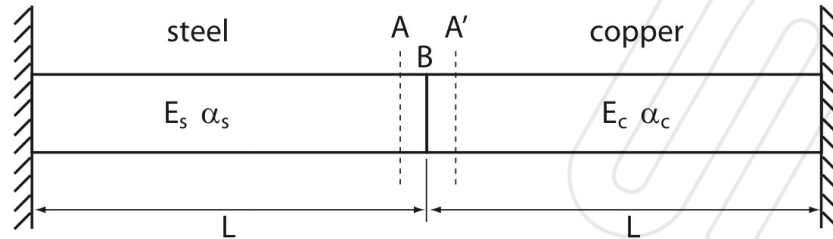
$$\therefore \sigma_{\max} = \frac{F}{A_{\min}} = 746.04 \text{ MPa}$$

$$(b) \quad \bar{\epsilon} = \frac{1}{3} \int_2^5 \frac{13500}{\pi \left( \frac{1}{2} \frac{3}{2500} x^2 \right)^2} dx = 1.5518 \times 10^8 \text{ Pa}$$

$$\therefore \epsilon = \frac{\bar{\epsilon}}{E} = 1.304 \times 10^{-3}$$

$$\therefore \Delta L = \epsilon L = 3.912 \times 10^{-3} \text{ m} = 3.912 \text{ mm}$$

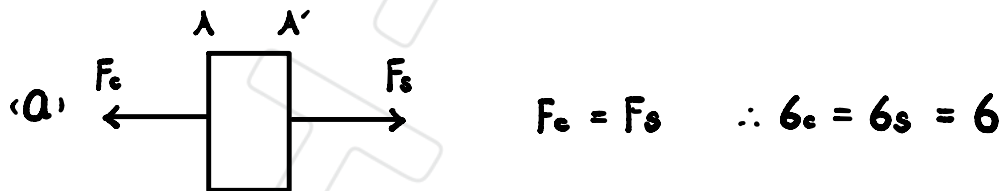
(4.) A steel rod and a copper rod with the same cross-sectional area are placed in contact at  $B$ . Both rods are constrained at their other end by rigid walls. The temperature is raised by an amount  $\Delta T$ , which causes the rods to be in a state of direct stress in the axial direction.  $E_s$  and  $E_c$ , and  $\alpha_s$  and  $\alpha_c$  are the Young's moduli and CTEs of steel and copper respectively.



- By drawing a free body diagram of the section between  $A$  and  $A'$ , write the equilibrium equation, and hence determine the stress.
- Assume Hooke's law applies and write the constitutive equations relating the stress to the strain in both rods.
- By considering the compatibility of the rod deformations, determine a relationship between the strains in the two rods. Hint What is the total extension?
- Using your results from a), b) and c), show that the magnitude of the stress in each rod is given by

$$|\sigma| = \frac{(\alpha_s + \alpha_c)\Delta T}{\left(\frac{1}{E_s} + \frac{1}{E_c}\right)} \quad (1)$$

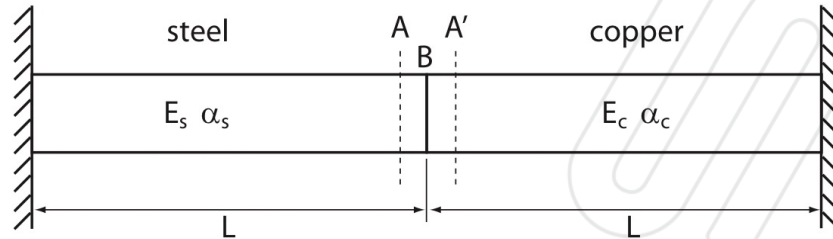
Is this stress tensile or compressive?



$$b) \quad \epsilon_c = \frac{\sigma_c}{E_c} + \alpha_c \Delta T = \frac{\sigma}{E_c} + \alpha_c \Delta T$$

$$\epsilon_s = \frac{\sigma_s}{E_s} + \alpha_s \Delta T = \frac{\sigma}{E_s} + \alpha_s \Delta T$$

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Is this stress tensile or compressive?

$$(c) \quad \epsilon_s = \frac{\Delta L}{L} = -\epsilon_c = -\left(\frac{-\Delta L}{L}\right)$$

$$(d) \quad \frac{\sigma}{E_c} + \alpha_c \Delta T = -\frac{\sigma}{E_s} - \alpha_s \Delta T$$

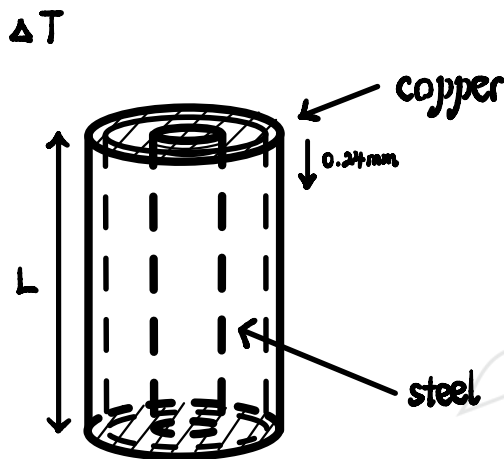
$$\therefore \sigma \left( \frac{1}{E_c} + \frac{1}{E_s} \right) = -(\alpha_s + \alpha_c) \Delta T$$

$$\therefore |\sigma| = \frac{(\alpha_s + \alpha_c) \Delta T}{\left(\frac{1}{E_c} + \frac{1}{E_s}\right)}$$

(5.) A steel rod 1.2 m long and 12 mm in diameter has screw-threaded ends and passes centrally through a copper tube with inner diameter 15 mm and outer diameter 25 mm. The ends of the tube are closed by rigid washers, which are secured by nuts on the screw-threaded ends of the rods. The nuts are tightened until the copper tube is reduced in length by 0.24 mm. The assembly is then heated resulting in a temperature rise of  $50^{\circ}\text{C}$ . Calculate the stresses in the steel rod and the copper tube before and after the temperature rise.

|        |                       |                                                 |
|--------|-----------------------|-------------------------------------------------|
| Steel  | $E = 207 \text{ GPa}$ | $\alpha = 11 \times 10^{-6}/^{\circ}\text{C}$   |
| Copper | $E = 103 \text{ GPa}$ | $\alpha = 17.5 \times 10^{-6}/^{\circ}\text{C}$ |

**Answer** Steel: 57.2 and 96.2 MPa (tension), Copper: 20.6 and 34.7 MPa (compression)



$$\text{Eq. } \sigma_s A_s + \sigma_c A_c = 0$$

$$\text{Comp. } \epsilon_s' = \frac{\Delta L_s}{L}$$

$$\epsilon_c' = \frac{\Delta L_c}{L} = -2 \times 10^{-4}$$

$$\epsilon = \epsilon_s = \epsilon_c$$

$$\text{Cons. } \epsilon_s = \frac{\sigma_s}{E_s} + \alpha_s \Delta T$$

$$\epsilon_c = \frac{\sigma_c}{E_c} + \alpha_c \Delta T$$

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**Answer** Steel: 57.2 and 96.2 MPa (tension), Copper: 20.6 and 34.7 MPa (compression)

$$\therefore \sigma_c' = \epsilon_c' E_c = -20.6 \text{ MPa}$$

$$\therefore \sigma_s' = \epsilon_s' E_s = -\frac{A_c}{A_s} \sigma_c' = -\frac{25}{9} \sigma_c' = 57.2 \text{ MPa}$$

$$\therefore \epsilon_c = \epsilon_s$$

$$\therefore \frac{\sigma_s}{E_s} + \alpha_s \Delta T = \frac{\sigma_c}{E_c} + \alpha_c \Delta T = -\frac{A_c}{A_s} \frac{\sigma_c}{E_s} + \alpha_s \Delta T$$

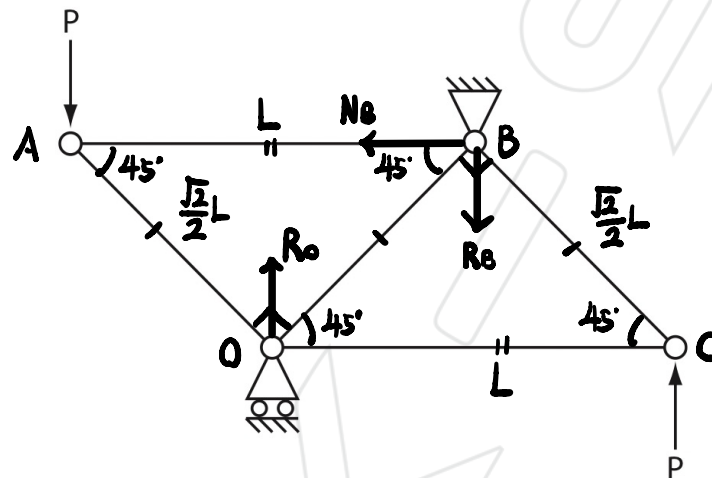
$$\therefore \sigma_c = \frac{(\alpha_s - \alpha_c) \Delta T}{\left(\frac{1}{E_c} + \frac{A_c}{A_s} \frac{1}{E_s}\right)} = -14.05 \text{ MPa}$$

$$\therefore \sigma_{ct} = \sigma_c' + \sigma_c = -34.7 \text{ MPa}$$

$$\therefore \sigma_{st} = \sigma_s' - \sigma_c \frac{A_c}{A_s} = 96.2 \text{ MPa}$$

## Statically-Determinate Truss Structures

(6.) In the structure shown, the horizontal bars have length  $L$ , and the diagonal bars are angled at  $45^\circ$  to the horizontal. Determine the support reactions and bar forces.



$$M = b + r - 2j = 5 + 3 - 2 \times 4 = 0$$

By inspection, infinitesimal mechanism

$$M(B), \quad \frac{1}{2}L \cdot P + L \cdot P = \frac{1}{2}L \cdot R_o \quad \therefore R_o = 3P$$

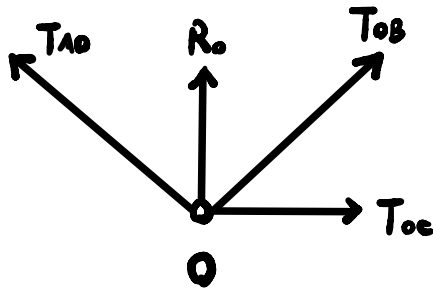
$$\therefore (\uparrow) P + R_b = R_o + P \quad \therefore R_b = 3P \quad \therefore (\rightarrow) N_b = 0$$

$$(\rightarrow) T_{AB} + \frac{\sqrt{2}}{2} T_{AO} = 0 \quad \therefore T_{AB} = -\frac{\sqrt{2}}{2} T_{AO}$$

$$(\uparrow) P + \frac{\sqrt{2}}{2} T_{AO} = 0 \quad \therefore P = -\frac{\sqrt{2}}{2} T_{AO}$$

$$\therefore T_{AO} = -\sqrt{2}P$$

$$\therefore T_{AB} = P$$



$$(\uparrow) \frac{\sqrt{2}}{2} T_{AO} + \frac{\sqrt{2}}{2} T_{OB} + R_O = 0 \quad \therefore T_{OB} = -2\sqrt{2}P$$

$$(\rightarrow) \frac{\sqrt{2}}{2} T_{AO} = \frac{\sqrt{2}}{2} T_{OB} + T_{OC}$$

$$\therefore T_{OC} = P$$

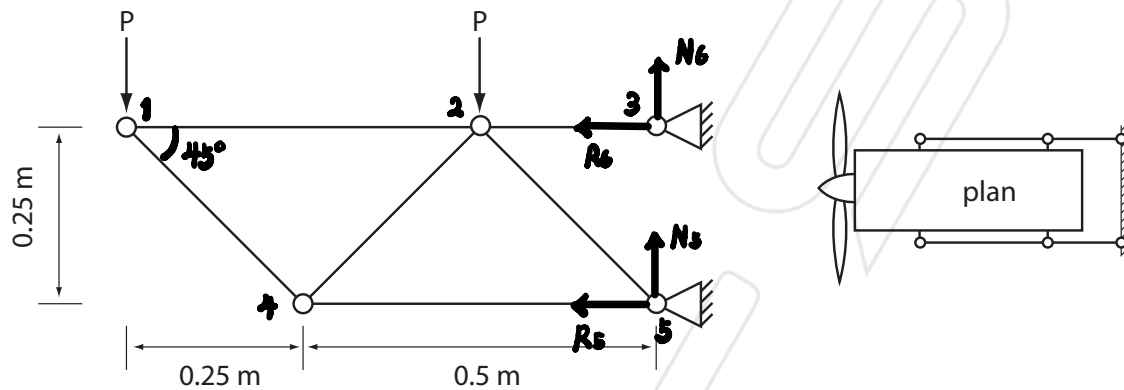
$$\therefore \text{Symmetry by OB} \quad \therefore T_{BC} = T_{AO} = -\sqrt{2}P$$

$$\therefore \text{Support reactions : } R_O = 3P, R_B = 3P$$

$$\therefore \text{bar forces : } T_{AB} = P, T_{AO} = -\sqrt{2}P,$$

$$T_{OB} = -2\sqrt{2}P, T_{OC} = P, T_{BC} = -\sqrt{2}P$$

(7.) One side of a small engine mount is represented by the truss shown in the figure. The engine weighs 250 kg and is attached to each support at two points where it transmits equal force  $P$ . The bars are made of Aluminium Alloy ( $E = 72 \text{ GPa}$ ). Determine the required cross-sectional areas of the bars if the maximum absolute value of the stress is to be kept below 200 MPa.



For these cross-sectional areas, calculate the bar extensions and associated direct strains. Comment on these strains.

$$M = b + r - 2j = 6 + 4 - 2 \times 5 = 0$$

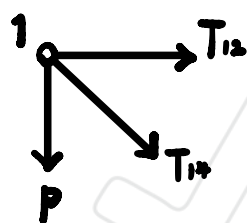
By inspection, statically determine

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$$(\uparrow) N_3 + N_6 = P + P = 2P$$

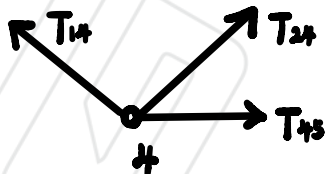
$$(\rightarrow) R_5 + R_6 = 0 \quad \therefore R_5 = -R_6$$

$$M(5), 0.25R_6 + 0.25P + 0.75P = 0 \quad \therefore R_6 = -4P \quad \therefore R_5 = 4P$$



$$(\uparrow) \frac{\sqrt{2}}{2} T_{14} + P = 0 \quad \therefore T_{14} = -\sqrt{2}P$$

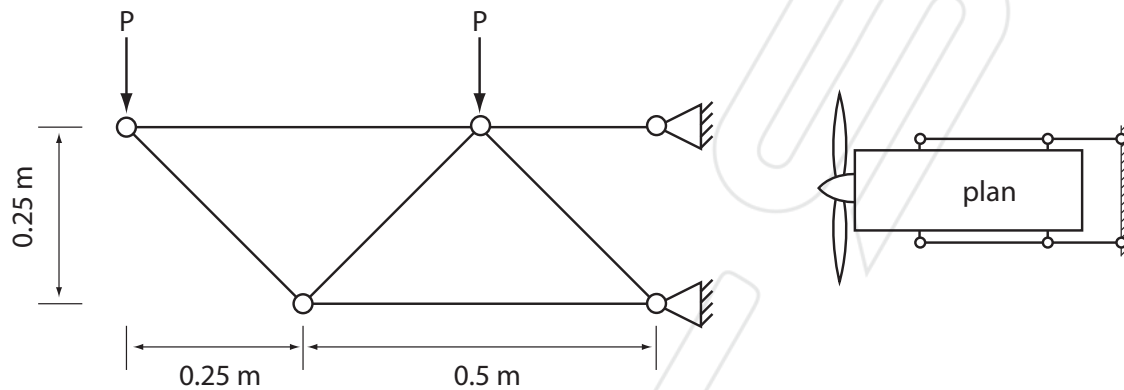
$$(\rightarrow) \frac{\sqrt{2}}{2} T_{14} + T_{12} = 0 \quad \therefore T_{12} = -\frac{\sqrt{2}}{2} T_{14} = P$$



$$(\uparrow) \frac{\sqrt{2}}{2} T_{14} + \frac{\sqrt{2}}{2} T_{24} = 0 \quad \therefore T_{24} = -T_{14} = \sqrt{2}P$$

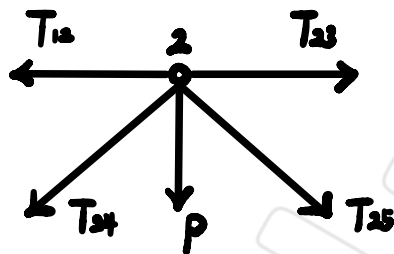
$$(\rightarrow) \frac{\sqrt{2}}{2} T_{14} = \frac{\sqrt{2}}{2} T_{24} + T_{45} \quad \therefore T_{45} = -2P$$

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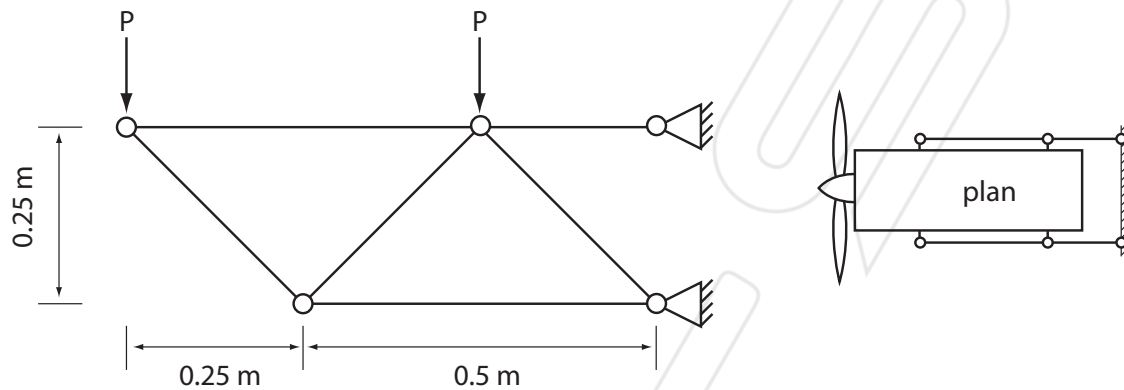
$$(\uparrow) \frac{\sqrt{2}}{2} T_{24} + P + \frac{\sqrt{2}}{2} T_{25} = 0 \quad \therefore T_{25} = -2\sqrt{2}P$$

$$(\rightarrow) T_{12} + \frac{\sqrt{2}}{2} T_{24} = T_{23} + \frac{\sqrt{2}}{2} T_{25} \quad \therefore T_{23} = 4P$$

$$\therefore 2 \times (2P) = W = mg$$

$$\therefore P = 613.125 \text{ N}$$

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$$\therefore \epsilon = \frac{\sigma_{\max}}{E} = \frac{1}{360}$$

$$\therefore \text{For } T_{12} = P, A_{\max} = 3.066 \times 10^{-6} \text{ m}^2, e = 1.37 \times 10^{-3} \text{ m}$$

$$\text{For } T_{14} = -\sqrt{2}P, A_{\max} = 4.335 \times 10^{-6} \text{ m}^2, e = -0.94 \times 10^{-3} \text{ m}$$

$$\text{For } T_{24} = \sqrt{2}P, A_{\max} = 4.335 \times 10^{-6} \text{ m}^2, e = 0.94 \times 10^{-3} \text{ m}$$

$$\text{For } T_{23} = 4P, A_{\max} = 1.226 \times 10^{-6} \text{ m}^2, e = 0.64 \times 10^{-3} \text{ m}$$

$$\text{For } T_{25} = -2\sqrt{2}P, A_{\max} = 8.671 \times 10^{-6} \text{ m}^2, e = -0.98 \times 10^{-3} \text{ m}$$

$$\text{For } T_{45} = -2P, A_{\max} = 6.131 \times 10^{-6} \text{ m}^2, e = -1.39 \times 10^{-3} \text{ m}$$